

The **SUBTOTAL** function in Excel is a versatile tool used to perform calculations like sums, averages, or counts on a filtered list or a specific range. Unlike standard functions, SUBTOTAL is unique because it can ignore rows that have been hidden by a filter.

1. Core Syntax

The function requires two main arguments:

`=SUBTOTAL(function_num, ref1, [ref2], ...)`

- **function_num**: A number (1 to 11 or 101 to 111) that specifies which calculation to perform (e.g., Sum, Average, Count).
- **ref1, ref2**: The range or ranges of cells you want to subtotal.

2. The "Hidden Row" Distinction

The `function_num` determines how the function treats hidden data:

- **Numbers 1–11**: Includes rows that you have **manually hidden** (right-click > Hide), but excludes rows hidden by an **AutoFilter**.
- **Numbers 101–111**: Excludes **all** hidden rows, whether they were hidden manually or by a filter.

Common Function Numbers

Calculation	Include Manual Hidden (1–11)	Exclude All Hidden (101–111)
AVERAGE	1	101
COUNT	2	102
MAX	4	104

SUM	9	109
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3. The Subtotal Feature (Data Tab)

Beyond the formula, Excel has a built-in **Subtotal command** located in the **Outline** group of the **Data** tab. This tool automatically inserts subtotal formulas and outline groups into a sorted list.

- **Automatic Grouping:** It identifies changes in a specific column (e.g., "At each change in: City") and inserts a subtotal row for that group.
- **Collapsible Outlines:** It creates "Level" buttons (1, 2, and 3) on the left side of the screen, allowing you to collapse the data to show only the grand totals or group totals.
- **Multiple Levels:** You can layer subtotals, such as showing the total salary by "Department" and then by "City".

4. Why use SUBTOTAL instead of SUM?

1. **Avoid Double Counting:** SUBTOTAL ignores other SUBTOTAL formulas within its range. If you have group subtotals and a grand total at the bottom, using `=SUBTOTAL(9, ...)` for the grand total will only add the original values, not the other subtotal results.
2. **Dynamic Filtering:** If you are analyzing "Sales data" and apply an **AutoFilter**, a standard SUM will still show the total of the entire list. Using SUBTOTAL(9, ...) will update to show only the total of the visible, filtered rows.
3. **Integration with Advanced Tools:** It works seamlessly with **Advanced Filter** and **Tables**, ensuring your dashboard metrics update as you narrow down your data.

5. Practical Workflow

1. **Sort Your Data:** Before using the Subtotal command, always sort your list by the column you want to group (e.g., Sort by "City" to group Hubli, Mysore, and Bangalore together).
2. **Apply Subtotal:** Go to **Data > Subtotal**, select your grouping column, and choose your function (e.g., Sum).
3. **Use Outlines:** Click the "2" in the top-left margin to hide the raw records and display only the summarized results for your report.